Markscheme

Mathematics: analysis and approaches
Practice question bank

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168. (a)

Consider the system $x + y + z = 6, 2x + 3y + kz = 14, x + 2y + 3z = 13$.

Find the value of k for which this system does not have a unique solution.

a unique solution fails when the determinant of the coefficient matrix is zero *(M1)*

$$\det = 1(9 - 2k) - 1(6 - k) + 1(4 - 3)$$ *(M1)* **A1**

$$= 4 - k$$ **A1**

set $4 - k = 0$: $k = 4$ **A1**

[6 marks]